DTU

DTU PCAS
Perception and Cognition
for Autonomous Systems

34759 Perception for Autonomous Systems

Final Project Description

Fall 2026 | DTU Electro

Project Description

At ASTA (Autonomous Systems Testing Arena), our robotics testing facility at DTU, our mobile robot, Husky, has been assigned to transport tools and equipment between testing stations. However, we suspect that Husky has been taking some rather unusual routes instead of going directly to its destinations. Luckily, two cameras have been recording its movements around ASTA, and we have obtained footage of four of its recent deliveries. Your task is to calibrate the cameras and develop a perception system that tracks Husky's movements and reconstructs its trajectory in 2D and 3D, even when it temporarily disappears from view.

What We Provide

- **Calibration images (0_calib):** Images of a calibration pattern used to calibrate the stereo camera system.
- **Sequence 1 (1_no_obstacle):** Raw (uncalibrated and unrectified) stereo image pairs of the Husky A200 robot driving at a constant speed and maintaining a constant distance from the camera, without occlusions.
- **Sequence 2 (2_obstacle):** Raw (uncalibrated and unrectified) stereo image pairs of the Husky A200 robot driving at a constant speed and maintaining a constant distance from the camera, with occlusions.
- **Sequence 3 (3_curved_bend):** Raw (uncalibrated and unrectified) stereo image pairs of the Husky A200 robot driving along a curved path, with its distance from the camera varying over time.
- **Sequence 4 (4_circle):** Raw (uncalibrated and unrectified) stereo image pairs of the Husky A200 robot driving along a circular path with its distance from the camera varying over time.

Project Goals

1. Calibrate the stereo camera system and rectify the input images.
2. Then, for each sequence, do the following:
    i. Compute disparity maps and reconstruct the 3D structure of the scene from the rectified stereo image pairs.
    ii. Detect the Husky A200 robot in the scene by processing the input images.
    iii. Track the Husky A200 robot over time in 2D and 3D, including estimating its position during complete occlusions.
3. Include a dedicated Discussion and Limitations section at the end of the report, reflecting on the overall approach, its limitations and failure cases, potential improvements, and possible directions for future work. Discuss how you would improve or approach the problem differently given more time or resources.

4. Present the results in a short video demonstration showing the detection and tracking of the Husky A200 robot across all four given sequences.

> **Important:** You are expected to solve the project using the **methods taught in the course**. You are welcome to explore deep learning approaches **if time permits**, but these should **complement**, rather than replace, the course methods.

Project Report

Each group must submit a single report documenting its final project at the end of the course. The report must be submitted via DTU Learn.

The report should explain and justify the chosen methods, describe the challenges encountered and how they were addressed, and present and discuss the results.

- **Length:** The report must be **10 $\pm$ 2 pages** long.
- **Video demonstration:** The report must include **a link to a video** demonstrating the group's results, preferably hosted on YouTube.

> **Evaluation:** The report is assessed on a pass/fail basis and accounts for 20% of the final course grade. A passing report contributes 20% toward the final grade, whereas a failing report contributes 0%. No intermediate grades are awarded (e.g., 10%).

Project Dates

Milestone	Date	Comment
Project release	2 November 2026	Material available on DTU Learn
Report submission	3 December 2026	Upload on DTU Learn